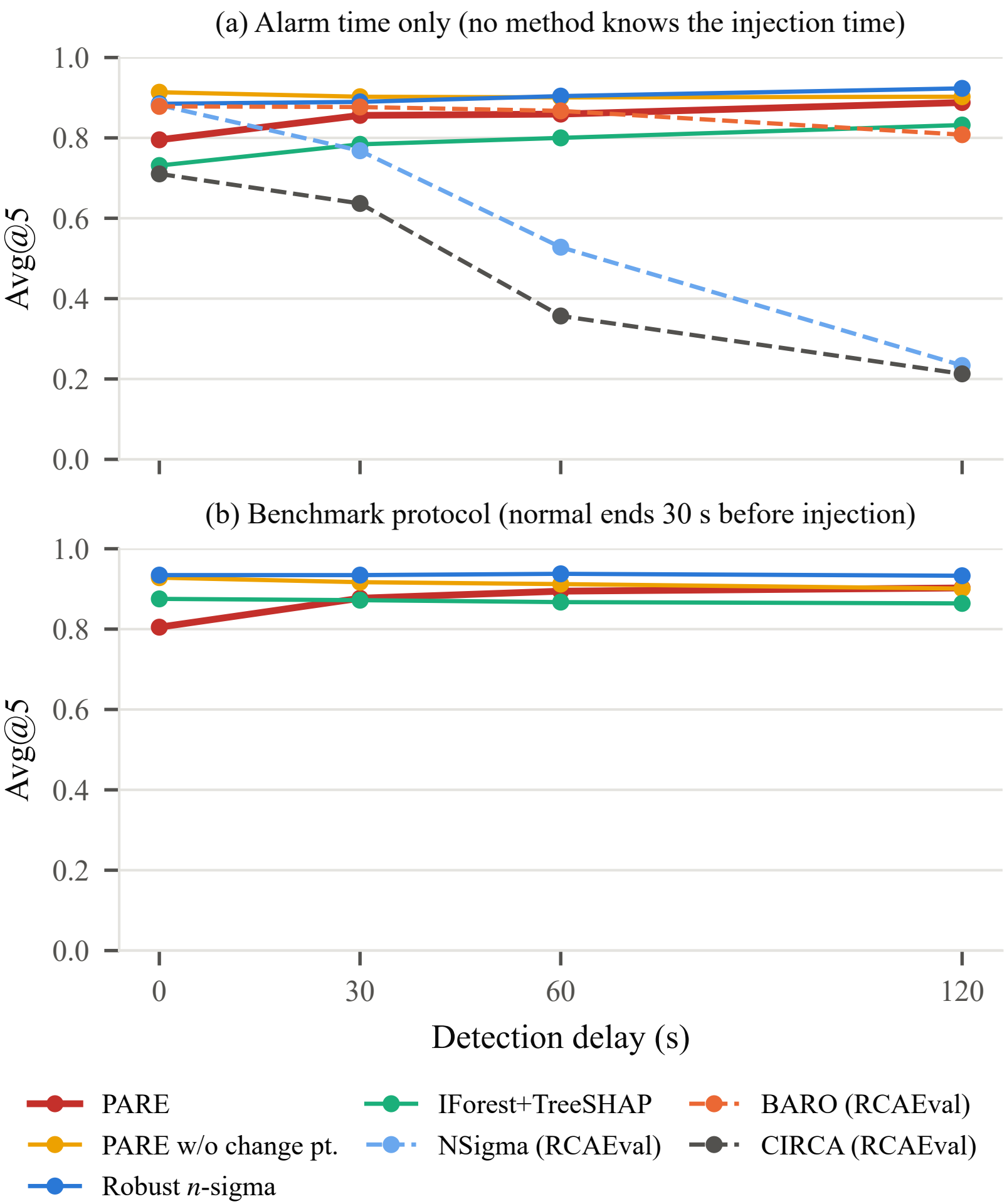
(a) Alarm time only (no method knows the injection time)
(b) Benchmark protocol (normal ends 30 s before injection)
Avg@5
1.0
0.8
0.6
0.4
0.2
0.0
0
30
60
120
Detection delay (s)
PARE
PARE w/o change pt.
Robust n-sigma
IForest+TreeSHAP
NSigma (RCAEval)
BARO (RCAEval)
CIRCA (RCAEval)